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Electrical power transmission to elevator system cars

Abstract

An elevator system includes a hoistway, an elevator car movable along the hoistway, and a power management and transfer system. The power management and transfer system includes an electrical power source, a hoistway contactor secured in the hoistway and operably connected to the electrical power source, and a car contactor disposed at the elevator car, such that when the car contactor is brought into operable contact with the hoistway contactor, electrical power is transferrable between the power source and the elevator car.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 62/779,550, filed Dec. 14, 2018, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The embodiments herein relate to elevator systems, and in particular to power transmission to elevator cars disposed in a hoistway of an elevator system.

(2) Conveyance systems, such as, for example, elevator systems, escalator systems, and moving walkways require electric power for operation. Travelling cables typically connect an elevator car of the elevator system to a stationary power source to provide power to the elevator car. Travelling cables may add expense, weight, and complexity to elevator car operation and installation.

SUMMARY

(3) In one embodiment, an elevator system includes a hoistway, an elevator car movable along the hoistway, and a power management and transfer system. The power management and transfer system includes an electrical power source, a hoistway contactor secured in the hoistway and operably connected to the electrical power source, and a car contactor disposed at the elevator car, such that when the car contactor is brought into operable contact with the hoistway contactor, electrical power is transferrable between the power source and the elevator car.

(4) Additionally or alternatively, in this or other embodiments the hoistway contactor is positioned at a hoistway wall of the hoistway at, a landing floor, or a hoistway ceiling.

(5) Additionally or alternatively, in this or other embodiments the hoistway contactor is positioned at a pit of the hoistway.

(6) Additionally or alternatively, in this or other embodiments the car contactor is positioned beneath a car floor of the elevator car.

- (7) Additionally or alternatively, in this or other embodiments a car position sensor is located in the hoistway, and a contactor drive is operably connected to the car position sensor and to the hoistway contactor. The hoistway contactor is movable from a retracted position into an extended position in operable contact with the car contactor by the contactor drive upon sensing of the presence of the elevator car by the car position sensor.
- (8) Additionally or alternatively, in this or other embodiments the contactor drive is one of an electrical motor or a linear actuator.
- (9) Additionally or alternatively, in this or other embodiments the contactor drive is configured to move the hoistway contactor from the extended position to the retracted position when the car position sensor does not sense the presence of the elevator car.
- (10) Additionally or alternatively, in this or other embodiments the contactor drive is configured to move the hoistway contactor from the extended position to the retracted position after a selected elapsed time at the extended position.
- (11) Additionally or alternatively, in this or other embodiments the car contactor is operably connected to an energy storage device disposed at the elevator car, the energy storage device charged via electrical power transfer from the power source.
- (12) Additionally or alternatively, in this or other embodiments the energy storage device is configured to provide electrical power to one or more elevator car electrical loads.
- (13) Additionally or alternatively, in this or other embodiments the one or more elevator car electrical loads are one or more of a lighting system, a ventilation system or a car drive system.
- (14) Additionally or alternatively, in this or other embodiments one or more of the car contactor or the hoistway contactor is formed from one of a metal leaf, roller or brush.
- (15) Additionally or alternatively, in this or other embodiments one or more alignment features align the hoistway contactor with the car contactor.
- (16) Additionally or alternatively, in this or other embodiments the one or more alignment features include one or more of a magnetic feature or a two-dimensional motor.
- (17) In another embodiment, a method of operating an elevator system includes moving an elevator car along a hoistway of an elevator system, operably connecting a hoistway contactor located at the hoistway to a car contactor disposed at the elevator car, and transferring electrical power between a power source and the elevator car via the operable connection of the hoistway contactor and the car contactor.
- (18) Additionally or alternatively, in this or other embodiments an energy storage device of the elevator car is charged via the transfer of electrical power from the power source.
- (19) Additionally or alternatively, in this or other embodiments one or more elevator car systems are powered by the energy storage device.
- (20) Additionally or alternatively, in this or other embodiments the hoistway contactor located at one of a landing floor or a pit of the hoistway is operably connected to the car contactor.
- (21) Additionally or alternatively, in this or other embodiments the presence of the elevator car is detected via a position sensor disposed in the hoistway, and the hoistway contactor is moved from a retracted position into operable connection with the car contactor when the presence of the elevator car is detected.
- (22) Additionally or alternatively, in this or other embodiments the hoistway contactor is returned to the retracted position when the presence of the elevator car is no longer detected.
- (23) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, that the following description and drawings are intended to be illustrative and explanatory in nature and non-limiting.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements.
- (2) FIG. 1 is a schematic illustration of an elevator system that may employ various embodiments of the present disclosure;
- (3) FIG. 2 is a schematic illustration of an embodiment of a power management and transfer system of an elevator system;
- (4) FIG. 3 is a schematic illustration of another embodiment of a power management and transfer system of an elevator system;
- (5) FIG. 4 is a schematic illustration of yet another embodiment of a power management and transfer system of an elevator system; and
- (6) FIG. 5 is a schematic illustration of still another embodiment of a power management and transfer system of an elevator system.

DETAILED DESCRIPTION

- (7) FIG. 1 is a perspective view of an elevator system **101** including an elevator car **103**, a counterweight **105**, a tension member **107**, a guide rail **109**, a machine **111**, a position reference system **113**, and a controller **115**. The elevator car **103** and counterweight **105** are connected to each other by the tension member **107**. The tension member **107** may include or be configured as, for example, ropes, steel cables, and/or coated-steel belts. The counterweight **105** is configured to balance a load of the elevator car **103** and is configured to facilitate movement of the elevator car **103** concurrently and in an opposite direction with respect to the counterweight **105** within an elevator hoistway **117** and along the guide rail **109**.
- (8) The tension member **107** engages the machine **111**, which is part of an overhead structure of the elevator system **101**. The machine **111** is configured to control movement between the elevator car **103** and the counterweight **105**. The position reference system **113** may be mounted on a fixed part at the top of the elevator hoistway **117**, such as on a support or guide rail, and may be configured to provide position signals related to a position of the elevator car **103** within the elevator hoistway **117**. In other embodiments, the position reference system **113** may be directly mounted to a moving component of the machine **111**, or may be located in other positions and/or configurations as known in the art. The position reference system **113** can be any device or mechanism for monitoring a position of an elevator car and/or counter weight, as known in the art. For example, without limitation, the position reference system **113** can be an encoder, sensor, or other system and can include velocity sensing, absolute position sensing, etc., as will be appreciated by those of skill in the art.
- (9) The controller **115** is located, as shown, in a controller room **121** of the elevator hoistway **117** and is configured to control the operation of the elevator system **101**, and particularly the elevator car **103**. For example, the controller **115** may provide drive signals to the machine **111** to control the acceleration, deceleration, leveling, stopping, etc. of the elevator car **103**. The controller **115** may also be configured to receive position signals from the position reference system **113** or any other desired position reference device. When moving up or down within the elevator hoistway **117** along guide rail **109**, the elevator car **103** may stop at one or more landings **125** as controlled by the controller **115**. Although shown in a controller room **121**, those of skill in the art will appreciate that the controller **115** can be located and/or configured in other locations or positions within the elevator system **101**. In one embodiment, the controller may be located remotely or in the cloud.
- (10) The machine **111** may include a motor or similar driving mechanism. In accordance with embodiments of the disclosure, the machine **111** is configured to include an electrically driven motor. The power supply for the motor may be any power source, including a power grid, which, in

combination with other components, is supplied to the motor. The machine **111** may include a traction sheave that imparts force to tension member **107** to move the elevator car **103** within elevator hoistway **117**.

(11) Although shown and described with a roping system including tension member **107**, elevator systems that employ other methods and mechanisms of moving an elevator car within an elevator hoistway may employ embodiments of the present disclosure. For example, embodiments may be employed in ropeless elevator systems using a linear motor to impart motion to an elevator car. Embodiments may also be employed in ropeless elevator systems using a hydraulic lift to impart motion to an elevator car. FIG. **1** is merely a non-limiting example presented for illustrative and explanatory purposes.

(12) In other embodiments, the system comprises a conveyance system that moves passengers between floors and/or along a single floor. Such conveyance systems may include escalators, people movers, etc. Accordingly, embodiments described herein are not limited to elevator systems, such as that shown in FIG. **1**.

(13) Referring now to FIG. **2**, the elevator system **101** includes a power management and transfer system **200**. An energy storage device **202**, such as a battery or the like, is located at the elevator car **103** and is utilized to power electrical systems of the elevator car **103**, for example, lighting system **204**, ventilation system **208**, or the like. In some embodiments, the energy storage device **202** may be utilized to provide electrical power to an elevator car **103** mounted drive system, such as a linear motor drive system **206**, or the like. An electrical power source **210** is located, for example, in the hoistway **117**, or alternatively outside the hoistway **117**.

(14) One or more hoistway contactors **212** are operably connected to the power source **210** and are located in the hoistway **117**. In some embodiments, such as shown in FIG. **2**, a hoistway contactor **212** is located at each landing floor **214** of the elevator system **101**, or at selected landing floors **214** of the elevator system **101**. A car contactor **216** is located at the elevator car **103**, and is connected to the energy storage device **202**. The car contactor **216** is configured and positioned such that when the elevator car **103** arrives at a landing floor **214** having a hoistway contactor **212** located thereat, the car contactor **216** aligns with the hoistway contactor **212**. Electrical power is then transmitted from the power source **210** to the energy storage device **202** via the alignment of the car contactor **216** with the hoistway contactor **212**. In some embodiments, the car contactor **216** and/or the hoistway contactor **212** are formed from, for example, a metal leaf, roller, or brush, which in some embodiments is spring loaded to ensure adequate contact between the car contactor **216** and the hoistway contactor **212** for transmission of electrical power therethrough to charge the energy storage device **202** of the elevator car **103**. While electrical power is transferred via the car contactor **216** and the hoistway contactor **212**, in other embodiments the contactors **212** and **216** may be utilized as a communications link to transfer data such as elevator system health information and load profiles between the elevator car **103** and the controller **115**. The communications link may be as an alternative to or in addition to the transfer of electrical power.

(15) While in the embodiment of FIG. **2**, the hoistway contactors **212** are located at the hoistway wall **218**, in other embodiments, other locations may be utilized. For example, in the embodiment of FIG. **3**, a hoistway contactor **212** is located in a pit **220** of the hoistway **117**, such that when the elevator car **103** is at its lowest position in the hoistway **117**, the hoistway contactor **212** aligns with the car contactor **216**, which in this embodiment is located beneath a car floor **222** of the elevator car **103**. Such an arrangement may be utilized in addition to or as an alternative to the configuration shown in FIG. **2**. Further, in some embodiments the hoistway contactor **212** may similarly be located above the elevator car **103** in which case the car contactor **216** aligns with the hoistway contactor **212** when the elevator car **103** is at its highest position in the hoistway **117**.

(16) Referring now to FIG. **4**, in some embodiments the hoistway contactor **212** is movable depending on the presence of an elevator car **103**. This alleviates a potential ride quality concern when compared to a fixed-position hoistway contactor **212** that the elevator car **103** will ride over

with each passing. As shown, the hoistway contactor **212** is mounted on a contact arm **224** and normally held close to the hoistway wall **218**. A position sensor **226** is located in the hoistway **117**. When the position sensor **226** detects the presence of an elevator car **103** at the hoistway contactor **212** location, the hoistway contactor **212** is extended via an arm drive **228** from its retracted position to an extended position such that the hoistway contactor **212** aligns with the car contactor **216** allowing for the transfer of electrical power therethrough. In some embodiments, such as shown in FIG. **4**, the arm drive **228** is an electrical motor or the like, and the contact arm **224** may take the form of a rotational actuator such as shown in FIG. **4**, or alternatively a linear actuator such as shown in FIG. **5**. One skilled in the art will readily appreciate, however, that these configurations are merely exemplary and that other arrangements may be utilized to move the hoistway contactor **212** into contact with the car contactor **216**. Once the position sensor **226** no longer detects the presence of the elevator car **103**, the arm drive **228** moves the hoistway contactor **212** to its retracted position. In other embodiments, the arm drive **228** moves the hoistway contactor **212** to its retracted position prior to movement of the elevator car **103**, for example, after a selected time period at the extended position. While in the embodiments described herein the hoistway contactor **212** is extended and retracted, one skilled in the art will appreciate that in other embodiments the car contactor **216** is extended and/or retracted for contact with the hoistway contactor **212**. Further still, in other embodiments both the car contactor **216** and the hoistway contactor **212** may be movable.

(17) In other embodiments, the power management and transfer system **200** may also include a mechanical release and latch system that could mechanically/kinematically cause the release and alignment of the hoistway contactor **212** and the car contactor **216** whereby the ability is controlled remotely. For example, an electromechanical control that releases an interface cam that allows the elevator car **103** motion to cause the rest of the mechanism to be aligned by the elevator car **103** itself. When not necessary the electromechanical device pulls the interface cam away from the elevator interface so that it can go through without interaction. In some embodiments, the power management and transfer system **200** may include one or more alignment features **300** to ensure alignment and sufficient contact between the car contactor **216** and the hoistway contactor **212**. Such features may include a magnetic feature and/or a 2D motor to correctly position the hoistway contactor **212** relative to the hoistway contactor **216**. In other embodiments, conic topologies of the car contactor **216** and the hoistway contactor **212** may be utilized. This way, there is a lot of clearance at the beginning of the interface but as there is closure of the gap, the conic section is guided into the proper orientation. The tips of the cones could be used for transferring current.

(18) The configurations disclosed herein provide a simple cost effective solution to provide electrical power to the elevator car **103**, specifically to charge the energy storage device **202**. This allows for the elimination of the typical travelling cable for electrical power transmission, improving ride quality and reducing cost of the elevator system **101**.

(19) The term “about” is intended to include the degree of error associated with measurement of the particular quantity and/or manufacturing tolerances based upon the equipment available at the time of filing the application.

(20) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(21) Those of skill in the art will appreciate that various example embodiments are shown and described herein, each having certain features in the particular embodiments, but the present disclosure is not thus limited. Rather, the present disclosure can be modified to incorporate any

number of variations, alterations, substitutions, combinations, sub-combinations, or equivalent arrangements not heretofore described, but which are commensurate with the scope of the present disclosure. Additionally, while various embodiments of the present disclosure have been described, it is to be understood that aspects of the present disclosure may include only some of the described embodiments. Accordingly, the present disclosure is not to be seen as limited by the foregoing description, but is only limited by the scope of the appended claims.

Claims

1. An elevator system, comprising: a hoistway; an elevator car movable along the hoistway; and a power management and transfer system including: an electrical power source; a hoistway contactor secured in the hoistway and operably connected to the electrical power source; and a car contactor disposed at the elevator car, such that when the car contactor is brought into operable contact with the hoistway contactor, electrical power is transferrable between the power source and the elevator car; a car position sensor disposed in the hoistway; and a contactor drive operable connected to the car position sensor and to the hoistway contactor; wherein the hoistway contactor is movable from a retracted position into an extended position in operable contact with the car contactor by the contactor drive upon sensing of the presence of the elevator car by the car position sensor.
2. The elevator system of claim 1, wherein the hoistway contactor is disposed at a hoistway wall of the hoistway at, a landing floor, or a hoistway ceiling.
3. The elevator system of claim 1, wherein the hoistway contactor is disposed at a pit of the hoistway.
4. The elevator system of claim 3, wherein the car contactor is disposed beneath a car floor of the elevator car.
5. The elevator system of claim 1, wherein the contactor drive is one of an electrical motor or a linear actuator.
6. The elevator system of claim 1, wherein the contactor drive is configured to move the hoistway contactor from the extended position to the retracted position when the car position sensor does not sense the presence of the elevator car.
7. The elevator system of claim 1, wherein the contactor drive is configured to move the hoistway contactor from the extended position to the retracted position after a selected elapsed time at the extended position.
8. The elevator system of claim 1, wherein the car contactor is operably connected to an energy storage device disposed at the elevator car, the energy storage device charged via electrical power transfer from the power source.
9. The elevator system of claim 8, wherein the energy storage device is configured to provide electrical power to one or more elevator car electrical loads.
10. The elevator system of claim 9 wherein the one or more elevator car electrical loads are one or more of a lighting system, a ventilation system or a car drive system.
11. The elevator system of claim 1, wherein one or more of the car contactor or the hoistway contactor is formed from one of a metal leaf, roller or brush.
12. The elevator system of claim 1, further comprising one or more alignment features to align the hoistway contactor with the car contactor.
13. The elevator system of claim 12, wherein the one or more alignment features include one or more of a magnetic feature or a two-dimensional motor.
14. A method of operating an elevator system, comprising: moving an elevator car along a hoistway of an elevator system; operably connecting a hoistway contactor disposed at the hoistway to a car contactor disposed at the elevator car; transferring electrical power between a power source and the elevator car via the operable connection of the hoistway contactor and the car contactor; detecting the presence of the elevator car via a position sensor disposed in the hoistway; and moving the

hoistway contactor from a retracted position into operable connection with the car contactor when the presence of the elevator car is detected.

15. The method of claim 14, further comprising charging an energy storage device of the elevator car via the transfer of electrical power from the power source.

16. The method of claim 15, further comprising powering one of more elevator car systems by the energy storage device.

17. The method of claim 14, further comprising operably connecting the hoistway contactor disposed at one of a landing floor or a pit of the hoistway to the car contactor.

18. The method of claim 14, further comprising returning the hoistway contactor to the retracted position when the presence of the elevator car is not detected.
